Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		6.45 cm^3		1		1		
		(ii)		moles thiosulfate = $0.500 \times \frac{6.45}{1000} = 3.225 \times 10^{-3}$ mol (1) reaction ratio ⇨ $2S_2O_3^{2-} \equiv 1I_2 \equiv 1\ ClO^-$ so moles chlorate(I) = 1.6125×10^{-3} mol (1) concentration chlorate(I) = $\frac{1.6125 \times 10^{-3}}{25 \times 10^{-3}} \times 10 = 0.645$ mol dm^{-3} (1) ecf possible from part (i)		3		3	2	
		(iii)		$0.645 \times \frac{74.5}{10} = 4.81\%$		1		1	1	
		(iv)		should have chosen 0.200 mol dm^{-3} marks credited for reasons, MAX 1 mark for reasons if a different concentration is chosen use a lower concentration so titration volume is greater – smaller percentage error / more accurate (1) cannot use too low a concentration / 0.0500 mol dm^{-3} as volume would be too large (for a standard burette) (1)			2	2		2
	(b)	(i)	I	order with respect to ClO_3^- ⇨ first order (1) order with respect to Br^- ⇨ first order (1) must show working to gain each mark		2		2	2	

Question				Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
			II	pH 1 has H^+ conc^n of 0.1 mol dm^{-3} / 10 times smaller than pH 0 (1) rate is 10^3 times lower rate = $\frac{9.18 \times 10^{-7}}{10^3}$ = 9.18×10^{-10} mol dm^{-3} s^{-1} (1)		1	1	2	2	
		(ii)		award (1) for each of following $e^{\frac{-52800}{308R}}$ = 1.0986×10^{-9} $e^{\frac{-52800}{298R}}$ = 5.4981×10^{-10} 1.0986×10^{-9} is double 5.4981×10^{-10} / $\frac{1.0986 \times 10^{-9}}{5.4981 \times 10^{-10}}$ = 2 therefore the rule is true under these conditions (1) or award (1) for each of following *k* at 298 = 5.1×10^{-6} *k* at 308 = 1.02×10^{-5} value of k at 308K is double its value at 298K therefore the rule is true under these conditions (1) ECF possible from incorrectly calculated values		1 1	1	3	3	

Question				Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		K_a of stronger acid would be <u>greater</u> than that for weaker acid as stronger acids have <u>greater dissociation</u>	1			1		
		(ii)	I	Joe's method: rearrangement of pV = nRT (1) n = 3.5 × 10^{-3} (1) accept alternative method Heledd's method: n = 3.48 × 10^{-3} or 3.484 × 10^{-3} (1) must show use of M_r of carbon dioxide to gain mark both answers to **appropriate** significant figures and both concentrations are the same (ECF from calculations) (1)		1 1 1	 1	4	3	1
			II	Heledd's method as measurements are to more significant figures / more precise / have higher resolution			1	1		1
			III	would make Heledd's experiment less accurate as the mass released would be much smaller (as H_2 has much smaller M_r than CO_2) (1) will not affect the accuracy of Joe's experiment as same volume of gas would be produced (1)			1 1	2		
		(iii)		pH from 2-6 (1) **must** attempt reason to gain this mark award (1) for either of following ammonium ion (partially) dissociates to release H^+ ions $NH_4^+ \rightleftharpoons NH_3 + H^+$ do not accept compound/salt/ammonium perchlorate releases H^+	1	1		2		
				Question 11 total	**2**	**14**	**8**	**24**	**13**	**4**